12th hacker series

Organism and populations

Ecology

Oikos → Habitat

Logos → study

organism ⇌ environment

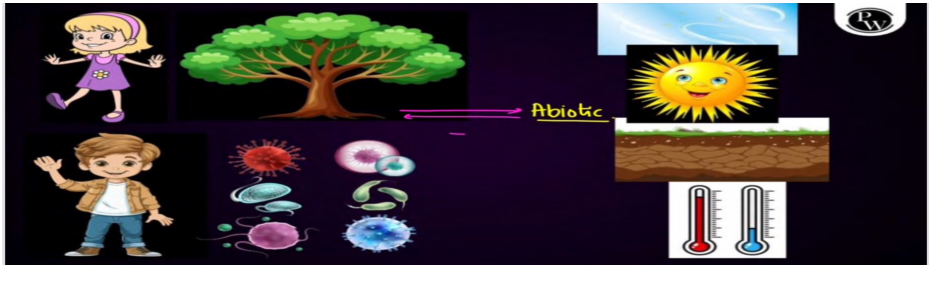
- ecology is the study of interaction among organism and interaction of organism with its environment

Biotic components

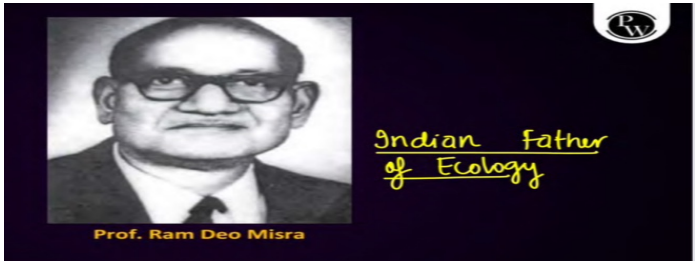
- plant
- Animals
- Microbes

Abiotic components

- water
- Light
- Soil
- Air
- Temperature



- term ecology was given by- Ernst Haechel
- Father of ecology- Alexander von Humboldt
- Indian father of ecology- Prof. Ram Deo Misra



Atoms → Molecules → Biomolecules → Sub-cellular components → Cell → Tissue → Organ → organ system → organism

Organism

- unit of ecology
- Unicellular/ multicellular
- Small/ big

Population

- group of organisms of same species in particular area

Community

- group of population in particular area is community
- Group of plant- plant community
- Group of animal population- animal community
- Group of microbial population- micro community

Ecosystem

- a single large ecosystem, minor variation
- Example- tropical rainforest, ocean

Biosphere

- largest ecosystem
- Global ecosystem
- Consist of atmosphere, hydrosphere, and lithosphere

population

- group of organisms
- of same species
- in a particular area
- Sharing or competing same resource
- Interbreed
- a sexually reproducing organisms are also included in population



Attributes of population

Organism level

- birth
- Death
- Sex (male, female)
- Age

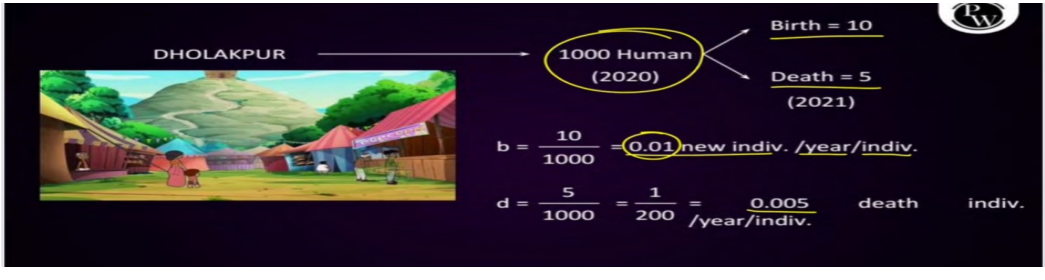
Population level

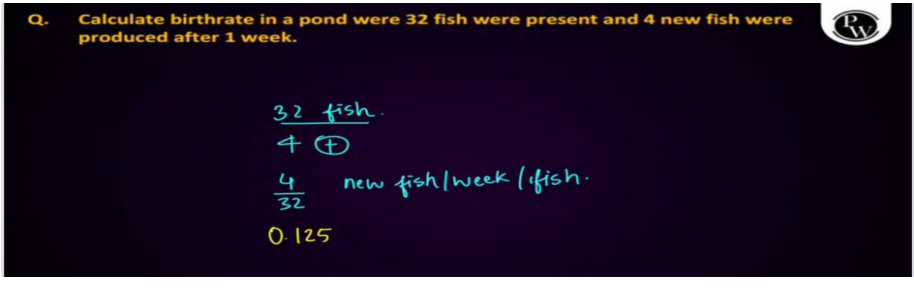
- birth rate
- Death rate
- Sex ratio
- Age distribution

Birth rate and death rate

- change a number of individual in member of population of particular area in given time
- is measured per capita
- Symbol - birth rate = b , death rate = d

$$b \text{ or } d = \frac{\text{change in no. of individual in given time}}{\text{Initial no. of individual}}$$





Sex ratio

- number of female per thousand male in particular area

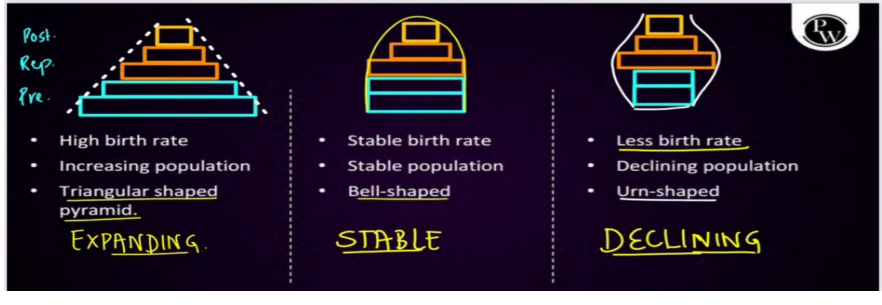
$$= \frac{\text{no. of female}}{\text{no. of male}}$$

Age distribution

- distribution of different age group categories studied with the help of age pyramid
- Three different categories
- 1) pre-reproductive
- 2) reproductive
- 3) post reproductive

Age pyramid

- describe age distribution in given area



Population density

- size of population in given area
- Symbol= N
- used to estimate
- 1) competition among species
- 2) effect of predation
- 3) effect of pesticides

Different method to calculate population density

- 1) total number
- 2) relative density
- 3) bio mass/ percent cover
- 4) pugmarks/ faecal matters

Total number

- example- 10, siberian cranes in Bharatpur, Rajasthan, millions of chlamydomonas in pond water
- to calculate total number is not possible, useless in many cases- example- fish in pond, bacteria in petridish dish

Relative density

- to calculate bacteria
- Fish in pond

Biomass /percent cover

- there are 200 plants of carrot grass
- One Banyan tree
- by mass one Banian is more than 200 carrot grass

Pugmarks/ feacal matter

- indirect census
- Used, calculate number of tiger ,lion etc

Population growth

- population is not stable
- Population is affected by
- 1) availability of food
- 2) effect of predation
- 3) weather condition

Four processes which affect population density

Mortality

- number of individual degree in a population in particular area in particular time of estimation

Natality

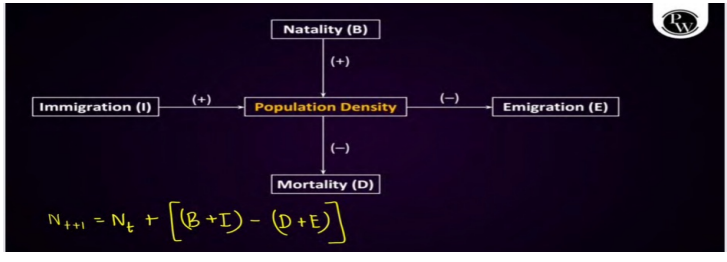
- number of individual increase in the population in particular area in particular time of estimation

Immigration

- number of individual come to particular area due to unfavourable condition

Emigration

- number of individual born from particular area due to unfavourable condition



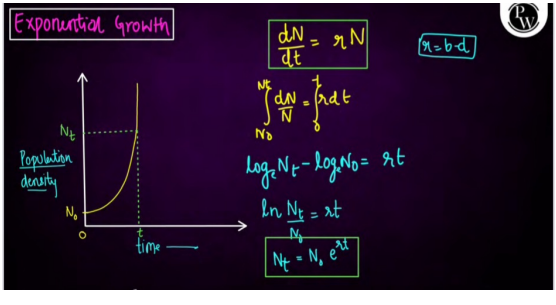
Population growth

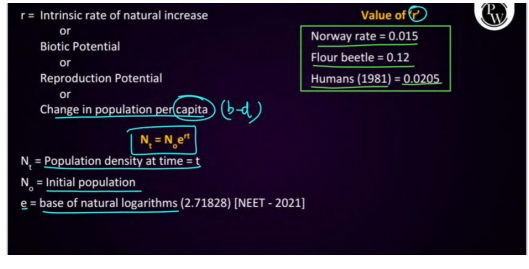
Unlimited resources

- exponential growth curve
- J shaped curve

Limited resource

- logistic growth curve
- Sigmoid curve

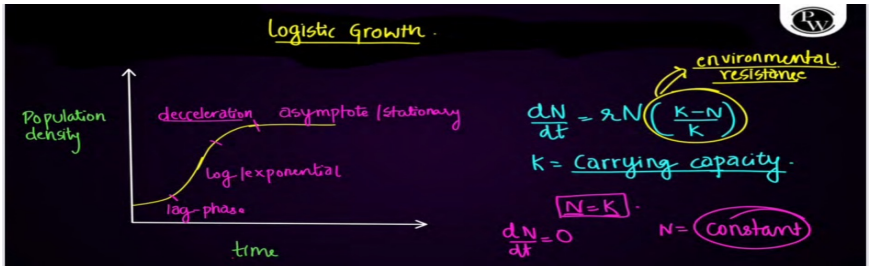


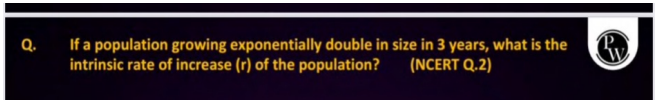


Exponential growth

- when resources are unlimited each species as the ability to realise fully its innate potential to increase a number



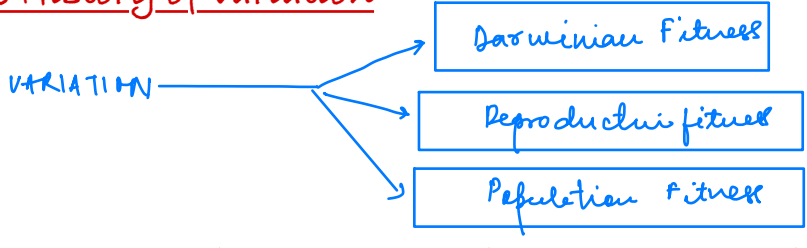




Logistic growth

- when resources are limited, population growth shows initially a lag phase followed by phase of acceleration and deacceleration, and finally asymptote
- it will result in sigmoid curve
- This type of population growth is called- verhulst pearl logistics growth

Life History of variation



- under a particular set of selection pressures, organism, evolve towards the most efficient reproductive strategy
- Time of reproduction → once in lifetime , eg - bamboo tree
  → Many times, eg - majority of mammals

- organism → reduce small size and large number of offspring, eg- oyster
  → Reduce large size and small number of offspring, eg - mammals

- ecologist suggest that life history traits of organism have you walked in relation to constraints, impose by abiotic and biotic components of the habitat

Population interaction

- there is no species which can survive in isolation
- Every plant require other microbes ,insects
- 1) for pollination
- 2)mycorhiza
- 3) for nitrogen fixation

- + = positive/ benefit
- - = negative/ loss
- 0 = neutral

Species A	Species B	Name of Interaction
+	+	Mutualism
–	–	Competition
+	–	Predation
+	–	Parasitism
+	0	Commensalism
–	0	Amensalism

Q. Which curve for growth rate is more realistic?

Mutualism

- this interaction confers benefit to both species

Mycorrhizae

- interaction of fungi and root of higher plants
- Both are benefited

Root

- provide shelter to fungi

Fungi

- increase surface area, more absorption of water and minerals
- Make soil resistant, drought resistant
- Protect from soil borne disease

Fig and wasp

- female wasp lay its egg into ovary of fig, This results in pollination of fig by wasp
- Which is pollinated by its partner wasp only and wasp use fig only
- it involves co- evolution
- Female wasp use fig not only for oviposition , also use its developing seed for nourishing it's Larvae
- they co- exist ,co-evolute and co-extinct

Yucca and moth

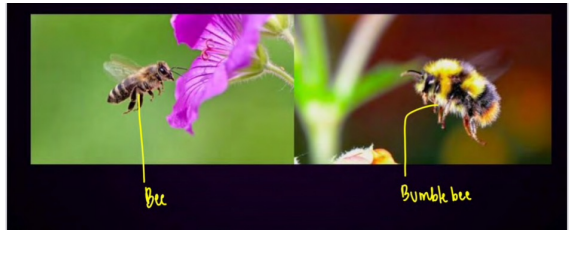
- yucca is pollinated by moth
- Yucca provide oviposition site
- moth Pollination by yucca plant

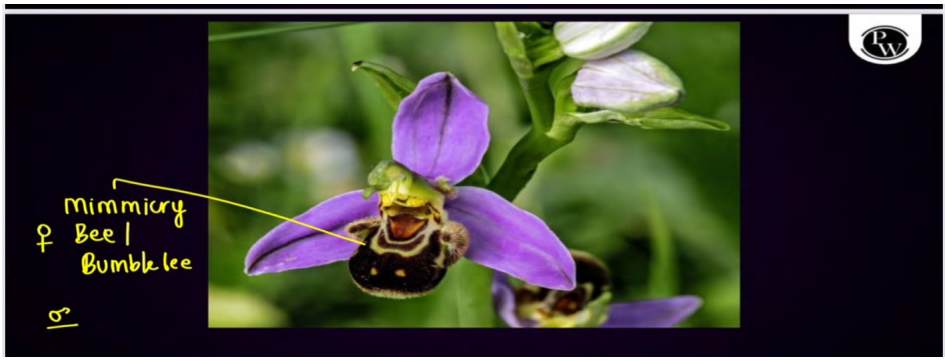
Cheaters/pollen nectar robbers

- try to steal Pollen and nectar, but do not help in pollination

Ophrys orchids and bee or bumble bee

- ophrys orchid is found in Mediterranean forest
- It's petal, show mimicry of female bee
- Male bee show pseudocopulation with petal and it is a sexual deceit
- petal Change itself according to change in female Bee thus it shows co-evolution





Lichens

- close association between fungi and algae
- Fungal component- it is called as mycobiont, it provides shelter to algae, absorption of water and mineral
- algal Component- it is called as phycobiont , it perform photosynthesis thus provide food

Commensalism

- in this interaction, one of the species is benefited and other one is neither harmed or benefited

1) epiphytes on mango branch

- orchids grow as epiphyte on mango branch, so as to come in exposure to sunlight to perform photosynthesis
- thus orchid is benefited and mango plant is neither benefited nor harmed

2) barnacle on whale

- Barnacle found on whale
- Benefits for barnacle- shelter, transportation, protection from predators
- Whale is neither harmed or benefited

3) cattle egrets and cattle

- when cattle move while grazing grass, small insects, come out of grass and egret bird eat those insects
- egrets benefits

4) sea anemone and clown fish

- plant fish is found in tentacles of sea anemone, thus it gets shelter and is protected from predators
- Clownfish benefits

Parasitism

- in this interaction, parasite feed on host
- Parasite= +
- host = -
- Small parasite, live on/ inside host body
- Get free lodging and free food

Adaptation

- loss of unnecessary sense organs
- Digestive system is absent
- High population density
- Presence of suckers to suck nutrition from host body

Effect of parasite on host

- reduce survival rate
- Reducegrowth and reproduction
- Reduce population density
- Make host more vulnerable to predators as it make host body weak
- Parasites are simple in morphology and internal structure, but have complex life-cycle
- Require more than one host to complete their life cycle

Co evolution

- if any change occurs in host, its parasite evolve itself accordingly

Good parasite

- which feed on host in proper way, does not harm host



Parasite

Endoparasite

→ Inside host

→ Ascaris
liver fluke
plasmodium
entamoeba histolytica

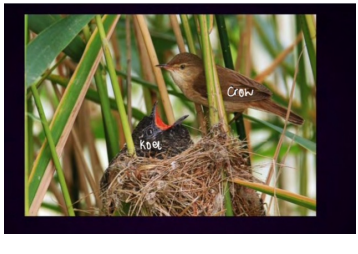
Ectoparasite

→ On host

→ Tick on dog
lice on human
copepods on fish

Brood parasitism

- special parasitism, which is not a true parasitism
- crow and cuckoo
- cuckoo Lay its egg in nest of crow
- Breeding season= spring to summer



Competition

- two closely related species compete for same resource
- Resources should be limited
- But these two conditions are not always true
- Totally unrelated species can also compare for same resources

- example- shallow South American lake
- Flamingo, and fish= compete for same resources
- Competition also occurs when resources are unlimited



Interference competition

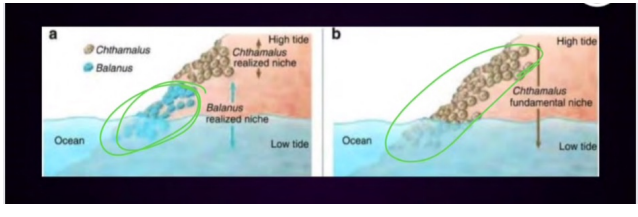
- feeding efficiency of members are affected because of presence of other species, even if resources are unlimited

Gause competition exclusive principle

- to closely related species, competing for same resource cannot coexist superior one will eliminate the inferior one
- Example- Galapagos island
- Abingdon tortoise- extinct within a decade, as browsing efficiency of goat was high

Competitive release

- inferior species has narrow distributional range, because of presence of competitively superior species
- if experimentally superior species were removed from that area, species, expand dramatically
- Eg- connell's elegant field experiment
- Superior species of balanus
- inferior species of chthamalus
- When superior species, ballanus was removed from low tide, competition release, chthamalus expanded

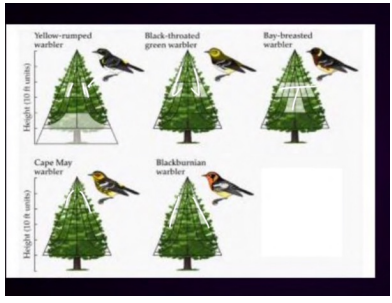


Resource partitioning

- closely related species can avoid competition by choosing
- Different feeding time
- Different pattern of foraging

Mac Arthur experience experiment

- five warbler bird, feed on same tree, to avoid competition, distribution of foraging pattern



Amensalism

- in this interaction, one of the species is harmed and other one is neither harmed or benefited
- The antibiotic released by fungi, kills bacteria
- Fungi, neither harmed or benefited

Predation

- in this interaction, one of the species is benefited and other one is harmed
- Catch, kill, eat
- Predator catch, kill, eat prey
- Even seed eating bird is predator
- Herbivores eating plants are predators

Significance of predator

- energy flow-

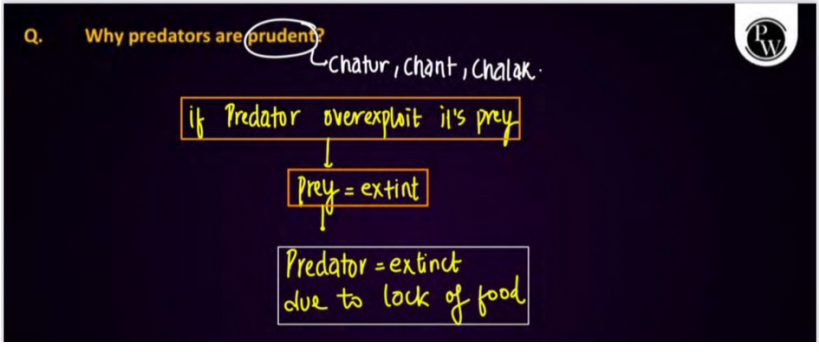
grass → Grasshopper → Frog → snake

predators act as conduit for energy transfer from one tropic level to next

- predators keep prey population under control- prickly pear cactus introduced in Australia in 1920, cause havoc in absence of natural predator, was controlled by natural predator (moth)
- Biocontrol agents- natural production is used to keep it's prey at manageable level

gambusia fish feed on mosquito larva and control mosquito population

- predators, maintain species diversity- by reducing competition among prey species, example- rocky intertidal area of Pacific coast of America, experimentally remove 10 species of invertebrates got extinct due to interspecific competition



Defence mechanism shown by prey





In Plants

1. Morphological Changes

Thorns → Bougainvillea, Acacia

Spines → Cactus, Opuntia





2. Calotropis

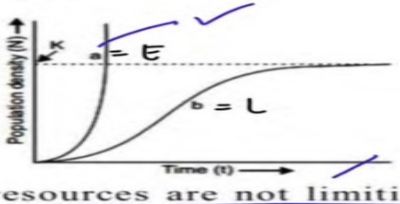
- Contain highly poisonous cardiac glycosides.
- Thus no browsing cattle / goat is found near it.
- 25% of insects are phytophagous (feed on plant species)

sap, other parts.





Select the correct option w.r.t. given population growth curve :



(1) a = resources are not limiting the growth

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$$

(2) b = resources are limiting the growth $\frac{dN}{dt} = rN$

(3) a = plot is exponential; $N_t = N_0 e^{rt}$

(4) b - plot is logistic ; $\frac{dN}{dt} rN\left(\frac{N-K}{K}\right)$

Correct option :

	Column I		Column II
(1)	Predation	(i)	Pseudocopulation
(2)	Mutualism	(ii)	Wasp Pollinator & fig inflorescence
(3)	Parasitism	(iii)	Fungal hyphea & Roots of higher plants as mycorrhiza
(4)	Ammensalism	(iv)	Cattle egret & grazing cattle

Lice on human, ticks on dogs, cuscuta on hedge plants, Lichen, Mycorrhiza & Rhizobium in Legume plants. How many are the example of parasitism?

(1) 3
(2) 5
(3) 4
(4) 6

Monarch butterfly is highly distasteful to its predator. It means :

(1) Prey & Predator are in close relationship
(2) Prey can prevent itself by predator
(3) Predator is a phytophagous species
(4) Predator can overexploit the prey

In the equation of logistic growth

$$\left[\frac{dN}{dt} = r_N\left(\frac{K-N}{K}\right)\right]\left(\frac{K-N}{K}\right)$$

(1) influence of environmental resistance
(2) intrinsic rate of natural increases
(3) exponential form of growth
(4) influence of individuals over biotic poten

Asymptote in a logistic growth curve is obtained when :

(1) $K > N$
(2) $K < N$
(3) The value of 'r' approaches zero
(4) $K = N$

K=N

A pyramid with pre-reproductive individuals being more than reproductive individuals the population is said to be .....a.... with ....b....age pyramid :

(1) a- Declining b- Urn shaped

(2) a- expanding b-Triangular

(3) a- stable b- bell shaped

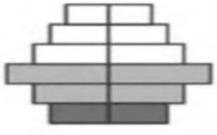
(4) a- expanding b- Bell shaped

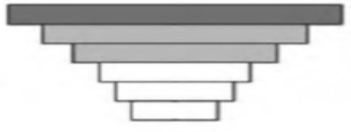
If '+' sign is assigned to beneficial interaction '–' sign to detrimental and 'O' sign to neutral interaction then the population interaction represented by '+', '–' refers to :

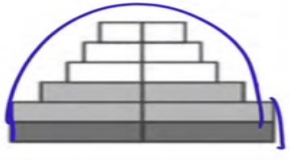
(1) Mutulasim ++ (2) Amensalism –O

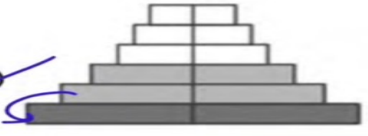
(3) Commensalism +O (4) Parasitism +–

Which of the given age pyramid reflects an increasing human population:

(1) 

(2) 

(3) 

(4) 

Resource partitioning mechanism *promotes*

(1) Co-existence

(2) Competitive exclusion

(3) Competitive release

(4) Amensalism

The principle of competitive exclusion was stated by :

(1) C. Darwin
(2) Gause
(3) Mac. Arthur
(4) Verhulst and pearl

Thank you!!